Video . 1

Intro to Linear Algebra

what is Linear Algebra

Algebra → Arithmetic (includes non numerical entities like x below:)

(x linear algebra)

$2x + 5 = 25$

$(2x+5) - 5 = 25 - 5$

$2x = 20$

$\boxed{x = 10}$

(If so, it has an exponential form then it isn't a linear algebra) → Eg

$2x^2 + 5 = 25$

$2\sqrt{x} + 5 = 25$

(Non Linear transformation)

(high definition of Linear Algebra)

Solving for the unknowns within a system of linear equation multiple

Eg

① Sheriff → 180 km/hr car

② Robber → 150 km/hr (5 min head start)

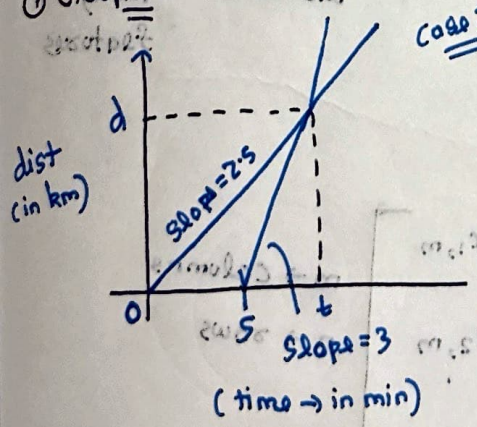
③ Q → time taken to catch the Robber

④ Q → Dist. travelled till that point

Solving

① Graphically

Case 1



(Note

150 km/hr → 2.5 km/min

180 km/hr → 3 km/min)

② Algebraically

Eqn 1 (Sheriff) ⇒ $d = 2.5t$

Eqn 2 (Robber) ⇒ $d = 3(t-5)$

So now,

$2.5t = 3(t-5)$

$2.5t = 3t - 15$

$2.5t - 3t = -15$

$-0.5t = -15$

$t = \frac{15}{0.5}$

$\boxed{t = 30 \text{ min}}$

$d = 2.5(30) = 75$ km

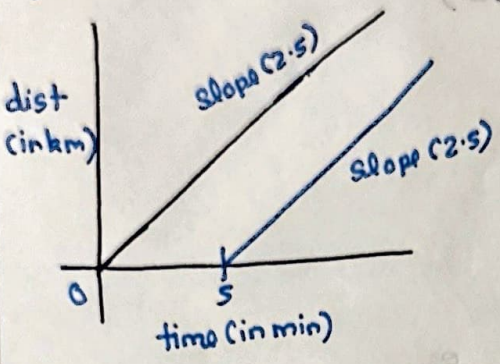
$d = 3(30-5) = 3(25) = 75$ km

Cases → Bank Robber

① if the BR and Sheriff travel with the same speed

BR → starts 5 min earlier

Case 2 (No Solution exist)



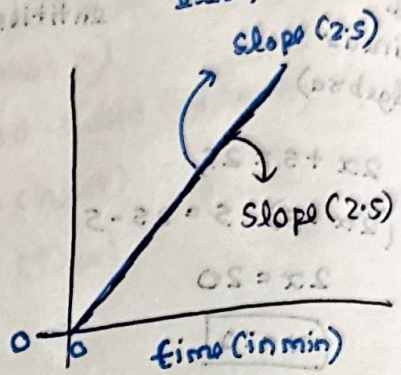
② if both of them start at

① same starting time

② same speed

(infinite solutions exist)

Case 3



So there are 3 options in Lin Alg →

① 1 solution

② no solution

③ infinite solution

In a given system, there could be

① many equations

② many unknowns in each equation

Eg → Regression model to predict house price

$$y = a + bx_1 + cx_2 + \cdots + mx_m$$

y → house price

a → y-intercept

x_1 → dist to school

x_2 → no. of bedrooms

x_m → could be many features

Regression model

$$y = a + bx_1 + cx_2 + \cdots + mx_m$$

$$\begin{bmatrix} y_1 & a + bx_{1,1} + cx_{1,2} + \cdots + mx_{1,m} \\ y_2 & a + bx_{2,1} + cx_{2,2} + \cdots + mx_{2,m} \\ \vdots & \vdots \\ y_n & a + bx_{n,1} + cx_{n,2} + \cdots + mx_{n,m} \end{bmatrix}$$

m → columns

n → rows

[for any house (i) in the dataset,

y_i = price and

$\boxed{x_i \rightarrow x_m}$ → features

we solve for parameters

a, b, c, ... m]